



Explosive mix

Italian chemist *Ascanio Sobrero* discovered **LIQUID NITROGLYCERIN** in 1847, but this explosive was so dangerous that even a knock could set it off. The **hazardous nature** of nitroglycerin made it impossible to use. Explosives needed to become **more stable** if they were ever to be beneficial.

How it changed

Dynamite transformed the construction industry, clearing the way for railroads and highways and blasting tunnels through rock.

the world

Dynamite

A truly **EXPLOSIVE** invention originally intended to put safety first

3. BANG! The shock from the exploding blasting cap makes the dynamite itself explode.

2. The blasting cap, a small explosive device attached to the dynamite, is set off by the current.

1. When the plunger is pushed into the blasting box, an electric current travels down the wire to the blasting cap.

BOOM!

Off with a bang!

After studying chemical engineering, Swedish scientist *Alfred Nobel* was determined to make more stable explosives. He found that mixing nitroglycerin with **KIESELGUHR** (a chalky sand) resulted in a safe explosive that could be dropped without exploding and yet **detonated on demand** with a blasting cap. He called his invention "**DYNAMITE.**"

Promise of peace

Nobel believed dynamite would **aid the construction** of roads, tunnels, and mines. However, to his disappointment, dynamite also became a **weapon of war**, so Nobel left his fortune to establish the **NOBEL PRIZE**. Since 1901, this annual set of awards recognizes the work that most helps humankind.

